Engineering Mechanics 1 · Unit_01 · UNIT 1

DRAFT — UE26ME142A · Unit_01 · UNIT 1 · UE26ME142A Unit 1 Class 10.1 AVSummary

Standalone video lesson · authored study notes with source citations

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· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

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M1

UE26ME142A Unit 1 Class 10.1 AVSummary

1 UE26ME142A Unit 1 Class 10.1 AVSummary

Study focus: UE26ME142A_Unit 1_Class 10.1_AVSummary. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000214 · EV0000215

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

My name is Dr.

Key details

- Sviven Kitish, Civil Engineering, Department of Science and Romantics, PS University.
- In the last class, we had discussed.
- few problems the unsolved problems from the prescribed textbook, Engineer Mechanics Statics by Miriam, Craig and Holton.
- Today is the thirteenth class, we'll be continuing to solve few more problems on the topic couple.
- The statement of the problem goes something like this.
- The system consisting of the bar OA.
- Two identical pulleys and a section of thin tape is subjected to the two 180 newton tensile forces shown in the figure, determine the equivalent force couple system at point O.
- There is a force of hundred and eighty newton force which is tensile force applied on a bar OA.
- This is bar OA.
- There are two pulleys, one is here, the other one is here.
- The force of 180 newton tensile is applied on these two pulleys.
- Which are both of them are horizontal, then he has asked us to determine the equivalent force couple system at point O.
- So equivalent force couple system at point O would be comprising of the net force because of these two forces plus the movement produced by these two forces at this point.
- Both of them are horizontal.
- The upper force is acting to the left, the lower force is acting to the right.

- You can look at that they form two equal unlike parallel force system, which is nothing but a couple.
- From the characteristics of a couple or properties of a couple that the net force acting because of a couple is zero.
- And that's what we will get here also.
- 180 newton force is acting to the left.
- The upper one, 180 newton force, the lower one is acting to the right.
- So to find out the force couple system at O, because of these two forces, there won't be any force.
- About if I take this force, I'll take the moment about this point.
- If I take this force, I'll take the moment about this point.
- That would be the moment at or at the center or at a point or anywhere you take, that would be the moment that is produced.
- Because it's a couple.
- This is horizontal, so I would like to know what is the vertical distance between the two forces, which is 180 and 180, these are the two forces.
- How do I get the vertical distance between these two forces?
- And we also know that OA is inclined at 45 degrees.

Example or application from the lecture

- Which has two identical pulleys, and if you observe the diagram that is given from this diagram, you can note on that.
- Then they have also given you what is the distance from O to the center of the lower pulley along the line OA, which is 50 mm, then the distance between the centers of the two pulleys along OA, which is 100 mm.
- He is also given the radius of the pulley as R and the value of R is also given as 25.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000214 · EV0000215 · EV0000216 · +1 source units in pack.json

Core explanation · 05:00–09:59

Sine 45 would get the vertical distance between the two centers of this pulley.

Key details

- We also know that this is radius, which is 25.
- Plus 25 will be the vertical distance.
- Which is acting in the anti-clockwise direction, which should be equal to the magnitude 180 newtons, cross or into the vertical distance between the two forces.
- And you can see the same thing here.
- This is the radius.
- This is also the radius.
- So obviously this would be the radius because this is the tangent, this would be acting tangent to this circle.

- You will get this as 21700 newton millimeters or 21.7 newton meters, which will be acting in the anti-clockwise direction, and we take anti-clockwise direction as positive.
- I hope all of you have understood this, right?
- We resort the force along the horizontal and multiply with the distance to get the movement at the pivotal point of the handle that is an O.
- Okay, because they are also a couple.
- Vertical distance, which is the perpendicular distance to these forces.
- I hope all of you have understood this, right?
- I think of this problem.
- This is a very important problem, and this problem.
- Which is 2 bar 78, the statement goes something like this.
- The force F acts along line MA, where M is the midpoint of the radius along the x-axis.
- Determine the equivalent force couple system at O if theta is equal to 40 degrees.
- From the statement, what we can know is there is a force that is acting.
- The line of action of the force is MA.
- The point M is the midpoint of the radius along the x axis, that is this is the x axis.
- Then they have also says that the angle theta is what it is.
- What is this angle theta?
- This is angle theta, which the radius OA makes with the x-axis.
- The force couple system of this force which is acting at this point, A, along M, a force couple system acting at so how do we solve it?
- Now I have to resolve this force first along horizontal and vertical.
- To resolve this force along horizontal and vertical, I should get this angle, say alpha.
- To get this angle alpha, this problem is majority of geometry and very less mechanics part is there.

Example or application from the lecture

- Who forces are acting parallel to each other on the sides or the edges of the handle of the two forces, there were acting parallel and horizontal distance was given.
- This is the radius and this M is the midpoint of this radius, and they have also given that the radius is R or they have given here this distance is R by 2, this is also R by 2, so the radius would be capital R.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000218 · EV0000219 · EV0000220 · +1 source units in pack.json

Core explanation · 10:00–14:59

I can find out what is the horizontal distance OB and the vertical distance AB.

Key details

- Because I know that this is R and this angle is theta.
- AB by M B.
- Or I can apply this force F here, which makes an angle of alpha with the x-axis which we have found.

- Next to determine the moment about O, I can take this force which is a component of F along x axis, which is F.
- Cos alpha multiplied with AB, so which will produce a clockwise moment which will be negative.
- The component of this force along y will be F sin alpha into this distance which is OB, which we have already found, which will produce an anti-clockwise moment which will be positive.
- This is just the arc.
- Then we have OA.
- Then we have.
- Now as I told you this angle is considered as alpha, which will be the same as this, because they are corresponding angles.
- I can get this to be equal to.
- I you can solve it by the approach that I have explained earlier, or by this approach, you take this triangle and use.
- The cosine rule.
- What does the cosine rule state?
- The side, any side, if I want to find out, a m that is this distance, a m square is equal to.
- This the other side R square and one more side plus R by 2 square minus 2 times R into R by 2 the other two sides.
- Cause of the angle opposite to the side on which we are we are supposed to find out that is AM.
- Opposite to AM is theta, that was that is what is mentioned here.
- I can find out alpha as 180 minus beta.
- This is another way of solving or determining the angle alpha.
- I have explained two different ways of solving alpha.
- You can use n number of ways to solve alpha.
- Determining of alpha is left to your knowledge that you have gained till now.
- You can solve it in n number of ways to determine alpha.
- Now once I get alpha by whatever means and method that I get, but.
- Make sure that the angle that you get is correct one, 67.5.
- Use correct logic to determine the angle alpha.
- Once you get angle alpha, now I will find out what is the moment about O by resolving this force along horizontal and vertical.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000222 · EV0000223 · EV0000224 · +1 source units in pack.json

Core explanation · 15:00–19:59

If I take this force to be acting at this point M itself, I can take this as F.

Key details

- Sin alpha into r by 2, that's what is taken here, which produces an anti-clockwise moment.

- Then I'm taking this as a point of application, so this component will be F cos alpha, F cos alpha, or I can also take the component of this by taking.
- Instead of alpha I'll take.
- Alpha is 65.
- M naught is equal to F sin alpha.
- This component, whatever you get cos alpha into there is no perpendicular distance because this four this component will pass through O itself.
- Mind it, what he has taken is F sin alpha is sine 67.5.
- He has not considered this component because it passes through O itself.
- Magnitude of F is same as F, the sense of the force will be the same as this, which is moving away from O or moving upwards.
- Then the direction of this force with respect to the x axis measured in the anti-clomage action positive x axis would be 67.5 plus the movement that's going to be acting at O will be equal to the value whatever we get from here which is 0.462 F into R.
- Earlier, what I explained was considering this as a point of application, I had explained how to get the moment about O.
- I had also explained how to get this angle alpha by taking or constructing a triangle.
- a b m and determining the distance mb and ab and then finding out alpha once you get alpha you take the point of application of this force as a itself determine the movement about O by resolving the forces of F along horizontal and vertical.
- Both the ways you can solve it.
- Of course, you can use n number of ways to solve this problem.
- That is, you can also resolve this force along this line.
- OA provided you can find out what is the angle made by the force of with respect to OA, and then resolving this along O and perpendicular to O A, then you will get F.
- component along this axis which is perpendicular to OA into this distance R, you will still get this answer provided you find out that angle whatever F makes with the line OA.
- And that is what we have done.
- The topic couple.
- This is the third topic under unit one, chapter two, we have discussed first topic was on resolution, rectangular, then projection, then N and T components.
- Then what is the difference between projection and component?
- Then the second topic was on moment.
- We have discussed how to determine moment, what is the positive and negative sign of moment.
- The third topic was couple.
- We have discussed couple characteristics of couple.
- The sense and classification of couples.
- Then we have discussed what is force couple system, and we have solved few problems.
- from the solid examples and few problems on from unsolved examples.
- In the next class that is class number fourteen, I'll start with the last topic of unit one that is resulted.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000226 · EV0000227 · EV0000228 · +1 source units in pack.json

Core explanation · 20:00–24:59

We have already discussed few things about resultant.

Key details

- I will discuss more about resultant in my next class.
- That is class number 14.
- My name is Dr.
- SV Venkage from the Department of Civil Engineering under Science and Humanities Department, PS University.
- See you in the next class.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000230 · EV0000231 · EV0000232 · +1 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
UE26ME142A_Unit 1_Class 10.1_AVSummary transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/010_UE26ME142A_Unit_1_Class_10.1_AVSummary__c39186c34db5b2f5.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 1_Class 10.1_AVSummary transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/010_UE26ME142A_Unit_1_Class_10.1_AVSummary__c39186c34db5b2f5.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
UE26ME142A_Unit 1_Class 10.1_AVSummary transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/010_UE26ME142A_Unit_1_Class_10.1_AVSummary__c39186c34db5b2f5.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9993; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 1_Class 10.1_AVSummary transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/010_UE26ME142A_Unit_1_Class_10.1_AVSummary__c39186c34db5b2f5.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. 2 rendered pages in 001_UE26ME142A_Unit_1_Class_1.0_Lecture_PPT__a53eb8b08a5ac7ff.slides.md have no readable `.slides.md` text; their original page images are included.
2. 2 rendered pages in 001_UE26ME142A_Unit_1_Class_1.1_Lecture_PPT__a6f76a088910b54a.slides.md have no readable `.slides.md` text; their original page images are included.
3. 2 rendered pages in 002_UE26ME142A_Unit_1_Class_2.0_Lecture_PPT__de74ea957f21118c.slides.md have no readable `.slides.md` text; their original page images are included.
4. 2 rendered pages in 002_UE26ME142A_Unit_1_Class_2.1_Lecture_PPT__bb478549c01cfb60.slides.md have no readable `.slides.md` text; their original page images are included.
5. 2 rendered pages in 002_UE26ME142A_Unit_1_Class_2.2_Lecture_PPT__3fc8761d12196c6f.slides.md have no readable `.slides.md` text; their original page images are included.
6. 3 rendered pages in 003_UE26ME142A_Unit_1_Class_3_Lecture_PPT__40166a1710a03d41.slides.md have no readable `.slides.md` text; their original page images are included.
7. 3 rendered pages in 004_UE26ME142A_Unit_1_Class_4_Lecture_PPT__984b7c368532af06.slides.md have no readable `.slides.md` text; their original page images are included.
8. 3 rendered pages in 005_UE26ME142A_Unit_1_Class_5_Lecture_PPT__f49d361d79853022.slides.md have no readable `.slides.md` text; their original page images are included.

9. 2 rendered pages in 006_UE26ME142A_Unit_1_Class_6_Lecture_PPT__f79618b4a0f260c2.slides.md have no readable `.slides.md` text; their original page images are included.
10. 3 rendered pages in 008_UE26ME142A_Unit_1_Class_8_Lecture_PPT__2659a840cc2bfd13.slides.md have no readable `.slides.md` text; their original page images are included.
11. 2 rendered pages in 009_UE26ME142A_Unit_1_Class_9_Lecture_PPT__832f697eaf1d383c.slides.md have no readable `.slides.md` text; their original page images are included.
12. 2 rendered pages in 010_UE26ME142A_Unit_1_Class_10.0_Lecture_PPT__aa8f331eb00d810f.slides.md have no readable `.slides.md` text; their original page images are included.
13. 2 rendered pages in 010_UE26ME142A_Unit_1_Class_10.1_Lecture_PPT__fe73ddc3ca596069.slides.md have no readable `.slides.md` text; their original page images are included.
14. 2 rendered pages in 011_UE26ME142A_Unit_1_Class_11_Lecture_PPT__3624142116c7273d.slides.md have no readable `.slides.md` text; their original page images are included.
15. 2 rendered pages in 012_UE26ME142A_Unit_1_Class_12.0_Lecture_PPT__2e7b61b526d51f60.slides.md have no readable `.slides.md` text; their original page images are included.
16. 4 rendered pages in 012_UE26ME142A_Unit_1_Class_12.1_Lecture_PPT__1d60317fe4af0cbe.slides.md have no readable `.slides.md` text; their original page images are included.
17. No historical archive question page met the unit-match candidate threshold; all subject-level and ambiguous pages remain in datasets/question-archive.json.
18. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
19. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
20. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
21. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00050	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0037. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0038 to s0054. Sidecar variants compare...
SRC00051	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0037. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0038 to s0054. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00052	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0037. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0038 to s0054. Sidecar variants compare...
SRC00053	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0037. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0038 to s0054. Sidecar variants compare...

C Revision and retrieval practice